

# Real Estate Price Estimation Using Machine Learning Models

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**Abstract**—Accurate real estate price estimation remains one of the important yet challenging problems in applied machine learning and urban analytics. Traditional appraisal methods are time-consuming, subjective, and often fail to capture complex non-linear relationships between property attributes and market prices. This study applies supervised machine learning models to predict residential property prices using the Melbourne Housing Full dataset, comprising 34,857 property transactions. After rigorous preprocessing—including IQR-based outlier capping, KNN imputation, feature engineering (such as *Age\_at\_Sale* and *Total\_Rooms*), label encoding, and log-transformation of the target variable—four regression models were trained and evaluated: Random Forest, Gradient Boosting, Lasso, and Ridge Regression. Random Forest emerged as the best-performing model, achieving an  $R^2$  of 0.7656 and a Mean Absolute Error (MAE) of \$174,244.75 on the test set, further validated by a 5-fold cross-validation mean  $R^2$  of 0.8240 with a standard deviation of 0.0063. Hyperparameter optimisation via GridSearchCV identified optimal parameters of  $n\_estimators=200$ ,  $max\_depth=20$ , and  $min\_samples\_split=5$ . Feature importance analysis confirmed that property *Type*, *Regionname*, and *Distance* from the central business district are the most influential predictors of sale price.

**Index Terms**—Real Estate, Machine Learning, Random Forest, Gradient Boosting, Feature Engineering, Automated Valuation Model, Melbourne Housing, Regression.

## I. INTRODUCTION

The global real estate market is among the most economically significant sectors in any country, with property prices shaped by a complex interplay of structural, locational, temporal, and socioeconomic factors. Reliable automated valuation of residential properties is critical for buyers, sellers, mortgage lenders, investors, and policymakers engaged in urban planning and tax assessment. Traditional appraisal methods—typically carried out by licensed valuers using comparative market analysis (CMA) or the hedonic pricing model—are inherently limited in scalability, reproducibility, and their capacity to process high-dimensional data [10]. A single appraiser cannot efficiently integrate dozens of structural and locational attributes simultaneously, nor can they adapt rapidly to shifting market conditions.

In recent years, machine learning (ML) has emerged as a powerful paradigm for automated valuation models (AVMs). ML algorithms can learn complex, non-linear mappings between property attributes and sale prices from large training datasets without requiring explicit feature engineering by domain experts

[7]. Ensemble tree-based methods such as Random Forest and Gradient Boosting have consistently demonstrated state-of-the-art performance across diverse real estate markets [4, 6]. Studies have shown that incorporating precise spatial data can improve forecasting accuracy by up to 52% [4], while engineered spatial features improved Random Forest classification accuracy by 19% in an Istanbul market study [8]. Beyond structural and spatial features, recent work has demonstrated that natural language property descriptions—processed via TF-IDF, Word2Vec, or BERT embeddings—can further enhance price prediction accuracy [3, 2, 5].

Despite the growing body of research, several gaps persist. Many existing studies focus on geographically narrow or luxury market segments [7], rely on asking prices rather than realised transaction prices [3, 6], or restrict their models to small, single-city datasets. The Melbourne residential property market, one of the most dynamic and internationally studied urban markets in the Asia-Pacific region, has received comparatively limited attention in terms of open-source, end-to-end ML pipeline documentation.

This study addresses these gaps through the following contributions: (1) a comprehensive preprocessing pipeline including IQR-based outlier capping, KNN imputation, and log-transformation of the target variable; (2) interpretable feature engineering producing *Age\_at\_Sale* and *Total\_Rooms* as composite predictors; (3) a systematic comparative evaluation of four regression models—Random Forest, Gradient Boosting, Lasso, and Ridge—using 5-fold cross-validation and GridSearchCV-based hyperparameter tuning; and (4) feature importance analysis identifying the primary drivers of residential property prices in Melbourne.

## II. LITERATURE REVIEW

A considerable number of studies have been conducted on predicting real estate prices using various feature sets and ML approaches. A more common direction involves combining structured property data with textual and spatial features.

Shen [1] used the Doc2Vec (Paragraph Vector) algorithm to project listing descriptions of 40,918 Atlanta single-family home transactions into a high-dimensional semantic space, computing pairwise cosine distances to yield a property “uniqueness score.” This score was integrated into a log-linear OLS hedonic model, demonstrating that a one standard deviation increase in uniqueness yields a price premium of approximately 5.6% (\$21,000). While the design prioritised causal inference, the linear OLS framework struggles with non-linear feature interactions and

remains geographically constrained.

Bittencourt et al. [2] conducted a rigorous benchmark of six text representation methods—Binary Bag-of-Words, Count BoW, Binary TF-IDF, Count TF-IDF, FastText, and fine-tuned BERT—compressed to 30 dimensions via truncated SVD and evaluated across five ensemble algorithms including Extra Randomised Trees (ERT), LightGBM, and XGBoost on dual-market data from Brazil and the USA. Simpler term-presence models such as Binary TF-IDF matched or outperformed BERT, with ERT emerging as the most effective algorithm. Text integration improved accuracy by up to 45% in the US market, though the study relied on asking rather than transaction prices.

Zhang et al. [3] compared TF-IDF, self-trained Word2Vec, and pre-trained BERT across Linear SVR, Random Forest, Gradient Boosting, and a three-layer DNN on 10,251 Canadian property listings. Gradient Boosting combined with TF-IDF achieved the best performance (RMSE = \$72,234 CAD), with self-trained Word2Vec substantially outperforming Google News pre-trained embeddings, underscoring the importance of domain adaptation in real estate NLP.

Tchuenté and Nyawa [4] evaluated seven ML models on 480,055 French real estate transactions (2015–2019) across nine metropolitan areas, comparing models built on standard structural attributes against those enriched with geocoded latitude-longitude coordinates. Random Forest achieved  $R^2$  up to 0.80 with MAE of approximately 29,000 euros; including geocoded data improved forecasting accuracy by up to 52%, confirming the primacy of spatial information in property valuation.

Ecevit et al. [8] reframed property valuation as a classification problem on 2,983 Istanbul listings, engineering six spatial features using Geopandas and Yandex APIs and training Decision Tree, Random Forest, and Logistic Regression classifiers via Apache Spark MLlib. Adding spatial features improved Random Forest accuracy by 19% to an overall 82%, demonstrating the power of relational spatial engineering over raw coordinates.

Alfiyatin et al. [9] combined hedonic linear regression with Particle Swarm Optimisation (PSO) on a time-series dataset of Indonesian house tax valuations in Malang, East Java. PSO was deployed to identify optimal regression coefficients rather than using ordinary least squares, reducing prediction error to IDR 14,186. While innovative in its metaheuristic optimisation approach, the study relied on government tax valuations rather than realised market prices.

Sankale and Hande [10] incorporated macroeconomic indicators—GDP, CPI, PPI, and HPI—alongside structural and geographic features, applying Decision Tree Regression and Support Vector Regression with target encoding. This holistic framing moves beyond purely property-specific attributes to account for market-cycle volatility, though detailed performance metrics were not fully reported.

Baldominos et al. [7] applied ExtraTreesRegressor, KNN, SVR, and MLP to 2,266 luxury property listings in Madrid’s Salamanca district. Ensemble regression trees achieved the best performance with a median absolute error of 94,850 euros (5.71%), while MLP performed poorly—a finding attributed to

the adverse effect of feature normalisation on neural network convergence.

Bushuyev et al. [5] integrated TF-IDF text features with numerical data for 9,260 Houston properties, using RFECV to select 38 key features and optimising a LightGBM model with Optuna, achieving  $R^2 = 0.834$ . SHAP TreeExplainer and Partial Dependence plots provided clear interpretability of individual feature contributions, including amenity attributes such as pools and elevators.

Kania Štykar [6] examined over 1.3 million Czech apartment advertisements (2014–2024), comparing Lasso, Ridge, Random Forest, and XGBoost combined with TF-IDF, Word2Vec, and BERT. Random Forest achieved the best performance ( $R^2 = 0.943$ , MAE = CZK 5,255), and textual data contributed more in rural settings where property descriptions carry comparatively richer unique information than in dense urban markets.

Across all reviewed studies, ensemble tree-based models—particularly Random Forest—consistently outperform linear and neural models on structured real estate data. Spatial and locational features dominate predictive power, log-transformation of price is widely recommended, and composite feature engineering measurably improves accuracy. Our work builds on these foundations in the context of the Melbourne residential market.

### III. METHODOLOGY

#### A. Dataset Description

The dataset used in this study is the *Melbourne Housing Full* dataset [11], a publicly available collection of residential property transaction records sourced from Kaggle. It contains 34,857 listings spanning multiple Melbourne suburbs with 21 raw feature columns. Key attributes include property *Type* (house “h”, unit “u”, or townhouse “t”), sale *Method* (auction, private sale, etc.), *Suburb* and *Regionname*, structural features including the number of *Rooms*, *Bathroom* count, *Car* spaces, *BuildingArea* ( $m^2$ ), and *Landsize* ( $m^2$ ), locational features including *Distance* from the Melbourne CBD (km) and *Propertycount* (number of properties in the suburb), and temporal information encoded in the sale *Date* and the year the property was built (*YearBuilt*). The target variable is the realised sale *Price* in Australian dollars.

Exploratory data analysis revealed that 21.8% of *Price* entries were missing (7,610 rows), representing the primary data quality challenge. Among structural features, *BuildingArea* had 56.0% missing values and *YearBuilt* had 38.8% missing. Distribution analysis confirmed strong right-skewness in *Price*, *Landsize*, and *BuildingArea*. Correlation analysis of numerical features indicated that *Rooms*, *BuildingArea*, and *Distance* held the strongest linear associations with *Price*.

#### B. Data Preprocessing

The dataset underwent a rigorous multi-step preprocessing pipeline to ensure data quality and compatibility with machine learning algorithms.

*Outlier Capping:* IQR-based capping was applied to *Landsize* and *BuildingArea*. For each column, the first quartile ( $Q_1$ ) and third quartile ( $Q_3$ ) were computed, and the interquartile range

$IQR = Q_3 - Q_1$  was used to define the acceptable range as  $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$ . Values outside this range were clipped using `DataFrame.clip()`, preserving all records while suppressing extreme outliers that would otherwise distort model training.

**Target Variable Filtering:** Rows with missing *Price* values were dropped, reducing the dataset from 34,857 to 27,247 records. This conservative strategy ensures that all retained records can be used directly for supervised regression without imputing the target.

**Column Removal:** Eight columns were dropped: *Address*, *Suburb*, *SellerG*, *Bedroom2*, *Postcode*, *CouncilArea*, *Latitude*, and *Longitude*. High-cardinality identifiers were excluded to prevent overfitting, *Bedroom2* was removed as a near-duplicate of *Rooms*, and raw coordinates were excluded in favour of the engineered proxy *Distance*.

**Median Imputation:** Missing values in *Landsize*, *BuildingArea*, and *YearBuilt* were filled with their respective column medians, a statistically robust strategy for right-skewed distributions.

**KNN Imputation:** Missing values in *Rooms*, *Bathroom*, *Car*, and *Distance* were imputed using *k*-Nearest Neighbours imputation ( $k = 5$ ), leveraging multivariate relationships among features to produce contextually appropriate fill values. Categorical columns *Regionname* and *Propertycount* were imputed with their respective modal values.

### C. Feature Engineering

Two new composite features were constructed to capture additional predictive information:

**Age\_at\_Sale:** The *Date* column was parsed to extract the transaction year, from which *YearBuilt* was subtracted to yield the property's age at the point of sale. Seven rows with negative ages (indicative of data entry errors in *YearBuilt*) were identified and clipped to zero. The original *Date*, *YearBuilt*, and *YearSold* columns were subsequently discarded. This engineered feature implicitly encodes structural depreciation as a price determinant.

**Total\_Rooms:** The sum of *Rooms* and *Bathroom* yielded a composite size indicator, after which the individual component columns were dropped to reduce collinearity while retaining combined information content. The final processed dataset comprised 27,247 records and 10 features (excluding *Price*).

### D. Label Encoding and Target Transformation

Categorical columns *Type*, *Method*, and *Regionname* were encoded using `sklearn's LabelEncoder`, converting nominal categories to integer representations. For linear models (Lasso and Ridge), all features were normalised using `StandardScaler` (zero mean, unit variance) to ensure gradient stability during optimisation.

The dataset was split 80/20 into training (21,797 samples) and testing (5,450 samples) subsets using `train_test_split` with `random_state=42`. Prior to model fitting, the target variable *Price* was log-transformed using `numpy's log1p` function to normalise its right-skewed distribution into an approximately Gaussian shape, improving regression model conver-

gence. Model predictions were subsequently back-transformed via `expm1` for evaluation in the original dollar scale.

### E. Learning Phase

Four supervised regression algorithms were trained and evaluated:

a) **Random Forest (RF):** Random forests build an ensemble of decision trees on bootstrapped subsets of training data, applying random feature subsampling at each split, and aggregate predictions by averaging across all trees [12]. This ensemble strategy substantially reduces variance compared to single decision trees, making RF robust to outliers and effective on mixed-type feature sets without requiring scaling. The initial model used `n_estimators=100` with `random_state=42`.

b) **Gradient Boosting (GB):** A sequential ensemble method that fits successive shallow decision trees to the residuals of prior iterations, progressively reducing prediction error [13]. Parameters were set to `n_estimators=100`, `learning_rate=0.1`, and `max_depth=3`. GB is often more accurate on tabular data but requires careful regularisation to prevent overfitting.

c) **Lasso Regression:** Ordinary least squares regression augmented with  $L_1$  regularisation (penalty coefficient  $\alpha = 0.001$ ), which encourages sparse solutions by shrinking uninformative feature coefficients to exactly zero [14]. Both linear models were trained on log-transformed prices using scaled features.

d) **Ridge Regression:** A linear model with  $L_2$  regularisation ( $\alpha = 1.0$ ), penalising the squared magnitude of all coefficients, which is particularly effective under multicollinearity. Ridge retains all features in the model, unlike Lasso.

The results after training and testing are discussed in the following section.

## IV. RESULTS AND DISCUSSION

### A. Feature Importance

We examined the correlation structure of the processed dataset to identify potential patterns and predictor dependencies. A correlation heatmap of the encoded numerical features was computed; however, as in prior studies [7], the heatmap provided limited directional insight into the relationship between individual features and sale price. Supplementary feature importance methods were therefore applied.

Feature importances extracted from the trained Random Forest model (Fig. 1) revealed that *Type*, *Regionname*, and *Distance* from the Melbourne CBD were the three most influential predictors of sale price. This finding is theoretically coherent: property type (house vs. unit vs. townhouse) encodes broad market segments with substantially different price distributions; regional classification captures macro-level spatial stratification in the Melbourne housing market; and CBD distance is a well-established negative predictor of residential price in monocentric urban economic models [4]. *BuildingArea* and the composite *Total\_Rooms* contributed moderately, while *Landsize*, *Car*, *Method*, and *Propertycount* exhibited comparatively lower importance scores.

A correlation heatmap after KNN imputation (Fig. 2) was also produced to examine inter-feature linear relationships, con-

Feature Importance — Random Forest Regressor

| Feature       | Importance |
|---------------|------------|
| Type          | 0.175      |
| Regionname    | 0.168      |
| Distance      | 0.152      |
| BuildingArea  | 0.114      |
| Total_Rooms   | 0.098      |
| Landsize      | 0.086      |
| Age_at_Sale   | 0.079      |
| Propertycount | 0.067      |
| Car           | 0.038      |
| Method        | 0.023      |

**Figure 1.** Feature Importance — Random Forest Regressor. Property Type, Regionname, and Distance from the CBD are the strongest predictors of residential sale price.

firming moderate positive correlation between *Total\_Rooms* and *Price*, and a moderate negative correlation between *Distance* and *Price*, consistent with the Random Forest importance ranking. PCA loading plots further corroborated these findings, with the first principal component capturing a strong combination of spatial (*Distance*, *Regionname*) and structural (*BuildingArea*, *Total\_Rooms*) variation, aligning with the dominant axes of price variability.

Correlation Matrix of Processed Features

[Heatmap: coolwarm palette, annotated with Pearson  $r$  values.

Strongest negative: *Distance-Price* ( $r \approx -0.38$ ).

Strongest positive: *Total\_Rooms-Price* ( $r \approx 0.44$ ).]

**Figure 2.** Correlation heatmap of all numerical features after preprocessing and imputation.

### B. Performance Metrics

Various machine learning models were evaluated to predict residential property prices from the processed dataset. Each model was assessed on two complementary metrics: Mean Absolute Error (MAE), which quantifies the average prediction error in the original dollar scale (Eq. 1), and the coefficient of determination ( $R^2$ ), measuring the proportion of variance in sale price explained by the model (Eq. 2).

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

Among the four models trained on log-transformed prices and evaluated after back-transformation to the original dollar scale,

Random Forest achieved the highest  $R^2$  of 0.7656 with an MAE of \$174,509. The other models demonstrated notably lower performance, with Gradient Boosting showing a moderate  $R^2$  of 0.6993 but a higher MAE of \$194,712. Both linear models (Lasso and Ridge) performed poorly, achieving only  $R^2 \approx 0.42$ , confirming the fundamentally non-linear nature of property price relationships.

**Table 1.** PERFORMANCE TABLE

| Model             | MAE (\$) | $R^2$ Score |
|-------------------|----------|-------------|
| Random Forest     | 174,509  | 0.7656      |
| Gradient Boosting | 194,712  | 0.6993      |
| Lasso             | 298,706  | 0.4185      |
| Ridge             | 298,684  | 0.4180      |

Analysing the performance metrics uncovers the clear superiority of ensemble tree-based models over linear approaches on this dataset. Random Forest explained 76.56% of sale price variance with an average absolute error of \$174,509—a reasonable margin across the wide Melbourne price range (\$85,000 to over \$9,000,000). Gradient Boosting achieved competitive accuracy but exhibited higher error on extreme-priced properties, suggesting greater sensitivity to residual outliers after log-transformation. The near-identical  $R^2$  scores for Lasso and Ridge, despite substantially different regularisation approaches ( $L_1$  vs.  $L_2$ ), indicate that neither sparsity nor coefficient shrinkage provides a meaningful advantage on this particular feature set, and that the limiting factor is the inherent non-linearity of the price-feature relationship.

To validate the robustness of the Random Forest model, 5-fold cross-validation was performed on the training set using log-transformed prices. The resulting  $R^2$  scores across the five folds were 0.8139, 0.8303, 0.8269, 0.8197, and 0.8295, yielding a mean of **0.8240** with a low standard deviation of 0.0063. This near-zero variance across folds confirms strong generalisation stability.

Hyperparameter optimisation via GridSearchCV over  $n\_estimators \in \{100, 200\}$ ,  $max\_depth \in \{\text{None}, 20, 30\}$ , and  $min\_samples\_split \in \{2, 5\}$  with 3-fold cross-validation identified the optimal configuration as  $n\_estimators=200$ ,  $max\_depth=20$ ,  $min\_samples\_split=5$ . The tuned model trained on 6 top features achieved MAE = \$179,845 and  $R^2 = 0.7549$ . Expanding to all 8 final features (*Type*, *Regionname*, *Distance*, *Total\_Rooms*, *Landsize*, *Propertycount*, *BuildingArea*, *Age\_at\_Sale*) yielded the best result: MAE = \$174,244.75 and  $R^2 = \mathbf{0.7656}$ , confirming incremental contributions from both structural size and temporal age information.

Overall, Random Forest with optimised hyperparameters performed best among all evaluated models and can thus be concluded to be the most accurate and robust predictor of Melbourne residential property prices in this study.

## V. CONCLUSION

Our models, despite certain limitations, managed to perform well in estimating real estate prices. This gives us a starting point and a clear vision of how to improve the pipeline to bring accuracy to a level suitable for deployment in professional automated valuation systems. Paying too much or too little for a property need not be an inevitable outcome for buyers and sellers. Our model aims to bring greater price transparency to the real estate market and relieve buyers and investors from uncertainty surrounding fair property values.

The Random Forest model emerged as the strongest performer with  $R^2 = 0.7656$  on unseen data and a cross-validation mean of 0.8240, substantially outperforming both Gradient Boosting and linear baselines. Feature importance analysis confirmed that property type, regional classification, and distance from the CBD are the dominant determinants of residential sale price, consistent with classical urban economic theory and prior ML literature [4, 7].

With further research, we aim to incorporate richer data sources—including property listing text descriptions processed via domain-fine-tuned BERT embeddings [2, 3], macroeconomic indicators such as interest rates and CPI [10], and neighbourhood amenity scores—to build a more comprehensive and accurate valuation model. Future work should also investigate one-hot or target encoding for categorical features to remove the false ordinality imposed by label encoding, and explore advanced frameworks such as XGBoost and LightGBM [5] with SHAP explainability to provide transparent, prescriptive pricing feedback to end-users.

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